Project Introduction

The **NHS England I02040 Compendium Emergency Readmissions Dataset** tracks the rate at which patients are readmitted to hospital as an emergency within 30 days of discharge. Covering **11 financial years from 2014/15 to 2024/25** across England, the dataset contains over **155,000 records** spanning national, regional, provider and clinical specialty levels. Emergency readmissions are a key NHS performance indicator high rates signal potential gaps in discharge planning, community care and patient support after leaving hospital.

Project Summary

- **Loaded and cleaned** the raw NHS Excel file in Power Query by removing the 15-row title block, promoting the correct headers and setting accurate data types for all numeric columns including Indicator value, Numerator and Denominator
- **Built a complete library of DAX measures** covering all KPIs national readmission rates, year-on-year changes, deprivation gaps, provider banding counts, rate ranges and dynamic badge labels with conditional colour coding for red, amber and green performance signals
- **Designed a multi-page interactive dashboard** with a full slicer bar controlling Year, Sex, Age and Breakdown featuring a national trend line chart, regional horizontal bar chart, deprivation gradient chart, provider banding donut chart and treatment function analysis all updating dynamically based on user selections
- **Applied advanced formatting** throughout including conditional bar colours highlighting the highest performing and worst performing areas, dashed England average reference lines on all charts, white data labels inside bars and a consistent NHS-aligned colour palette across all visuals
- **Embedded a Python visual** using Matplotlib to render a custom donut chart with dynamic centre text showing total provider counts and colour coded segments demonstrating how Python scripting inside Power BI can extend beyond native visuals to deliver publication quality outputs

Key Insights

1. Readmission Rates Have Been Rising Steadily for a Decade (Reference: National Trend Line Chart) The England readmission rate increased from 12.6% in 2014/15 to 14.8% in 2023/24, a 17% relative rise over 10 years. This upward trend is consistent across males, females and all persons, pointing to sustained and growing pressure on NHS discharge and community care pathways.

2. COVID-19 Created the Largest Single Year Spike on Record (Reference: YoY Change Bar Chart) The 2020/21 financial year saw the largest single year increase of +1.10pp driven by a sharp collapse in elective admissions during the pandemic. This was immediately followed by a correction of -1.00pp in 2021/22 as activity resumed. All other years show comparatively modest movements between +0.10pp and +0.60pp.

3. National Performance at a Glance (Reference: KPI Cards) In 2023/24 there were 939K emergency readmissions from 6.34M eligible admissions producing a national rate of 14.8%, up 0.6pp on the prior year. Of 308 providers assessed, 47% are performing better than expected, 37% within expected range and 15% are flagged as alert requiring urgent attention.

4. Regional Variation Spans 2.0 Percentage Points Across England (Reference: Regional Bar Chart) The South West records the highest readmission rate at 15.8%, sitting above the England average of 14.87%, while the East of England performs best at 13.8%. The South West, North East and South East all exceed the national average, suggesting geography and local health system capacity are significant drivers of performance differences.

5. Nearly Half of Hospital Providers Are Performing Better Than Expected (Reference: Provider Banding Donut Chart) Of 308 providers, 46% are banded as better than expected and 36% are within the expected range. However 18% are flagged as alert above expected, meaning approximately 1 in 6 hospitals has readmission rates that cannot be explained by patient complexity alone and warrant further investigation.

6. Deprivation Drives a Persistent 1.9pp Inequality Gap (Reference: Deprivation Bar Chart and Trend Line) The most deprived communities record a readmission rate of 15.0% compared to 13.1% for the least deprived, a gap of 1.9 percentage points. This gradient has remained consistent across all 11 years of data with no sign of narrowing, confirming that socioeconomic inequality is a structural driver of emergency readmissions rather than a temporary pattern.

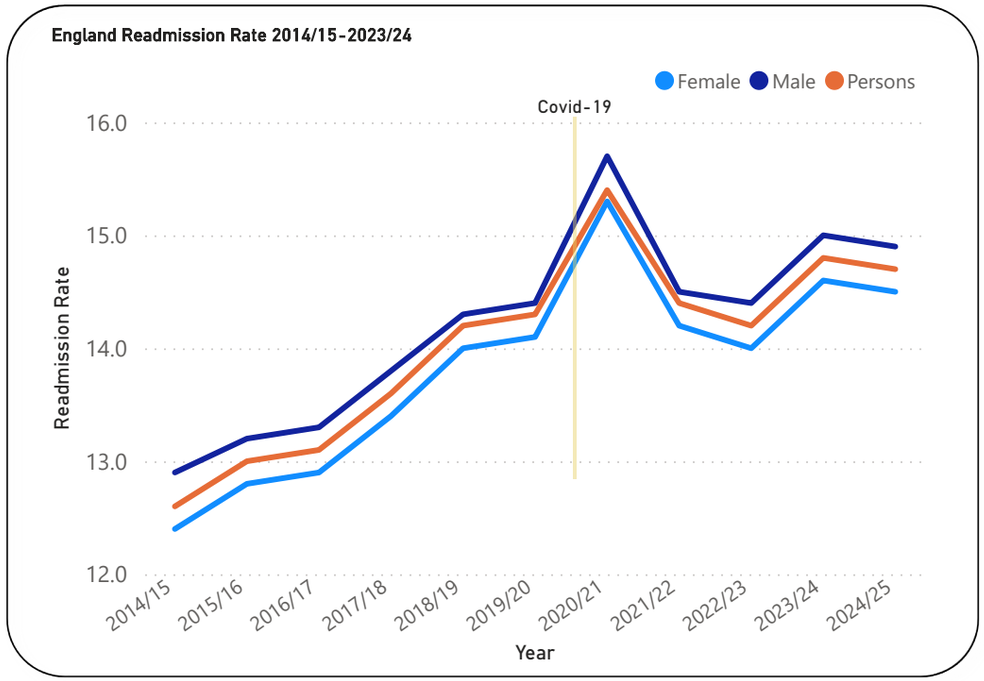
Overall Conclusion

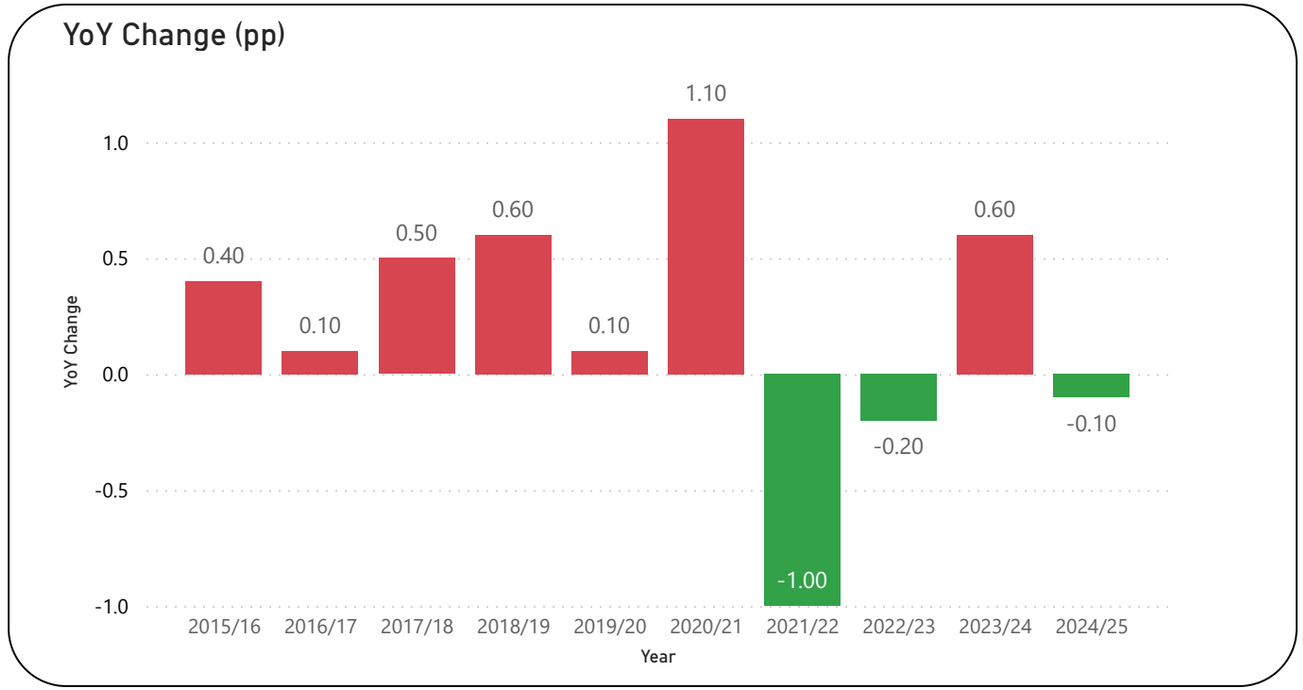
Emergency readmission rates in England are rising, geographically unequal and strongly linked to deprivation. The South West consistently underperforms the national average while the East of England shows the most favourable outcomes. A persistent 1.9pp deprivation gap confirms the most vulnerable communities carry a disproportionate readmission burden. With 1 in 6 hospitals flagged as alert and rates continuing to climb, targeted intervention in underperforming regions and deprived communities remains a clear priority for NHS leadership.

NHS England — Emergency Readmissions Dashboard

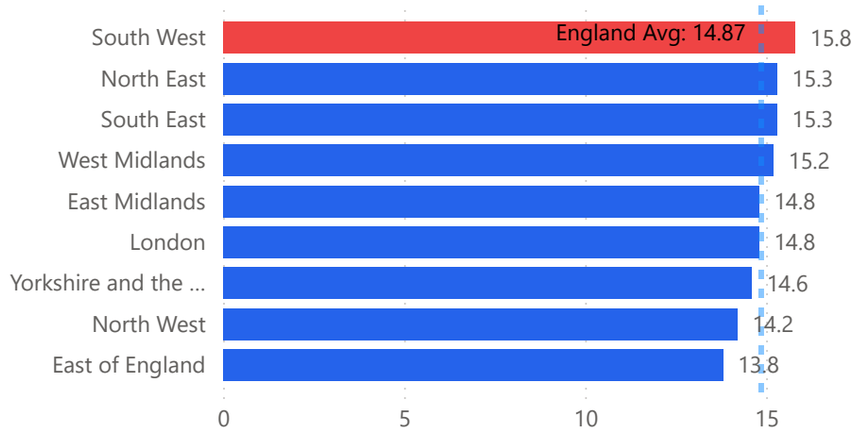
Dataset: I02040 Compendium **Coverage:** 01/04/2014 to 31/03/2015 **Records Loaded:** 155573

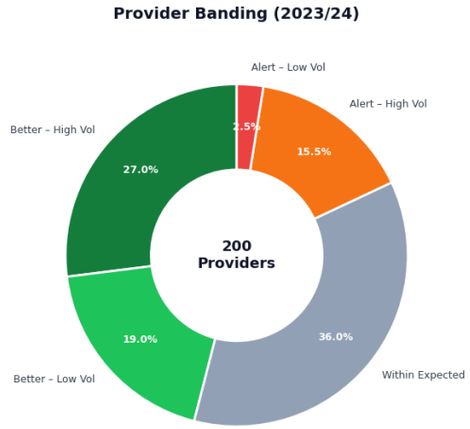
Total Readmission	Readmission Rate	Deprivation Gap (2023/24)	Provider Better than Expected	Provider Within Expected	Provider Alert > Expected
939K	14.8%	1.34pp	146	115	47
of 6.34M eligible	↑ +0.6pp vs 2022/23	Most vs Least Deprived	47% of 308	37% of 308	15% of 308



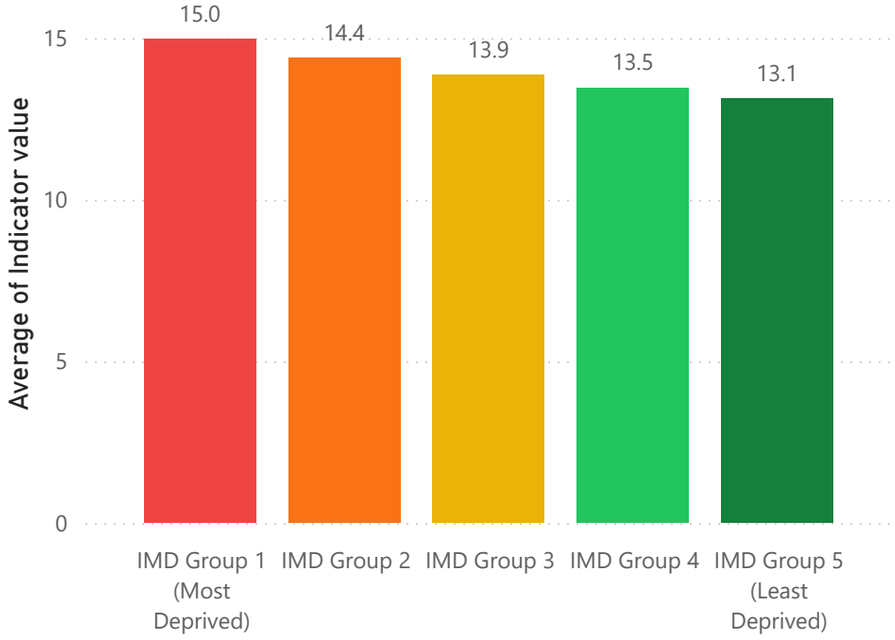
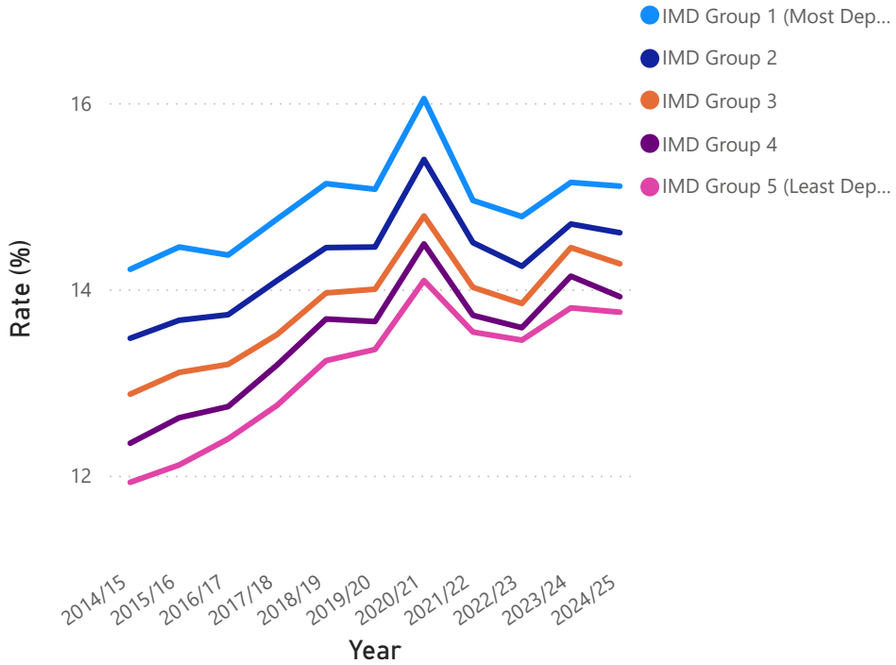


Region - Readmission Rate (2023/24)





Readmission Rate By Deprivation Quintile (2023/24)



Deprivation Trend Over Time